

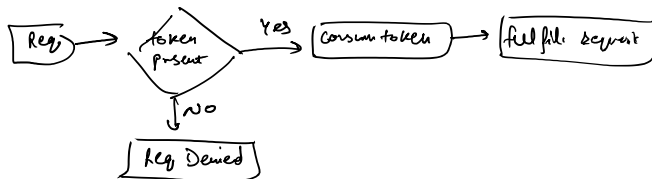
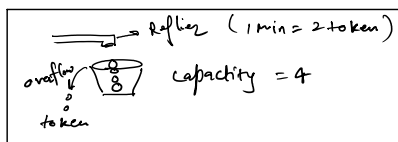
Design Rate Limiter

It is used to prevent DDOS Attack

we need to know 5 Algo

Token bucket Leaking bucket Fixed window counter Sliding window log Sliding window counter

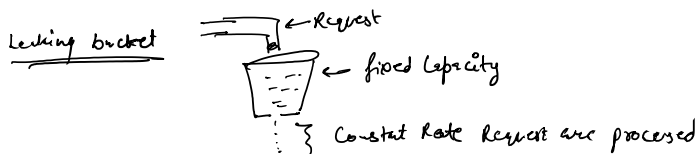
Token bucket



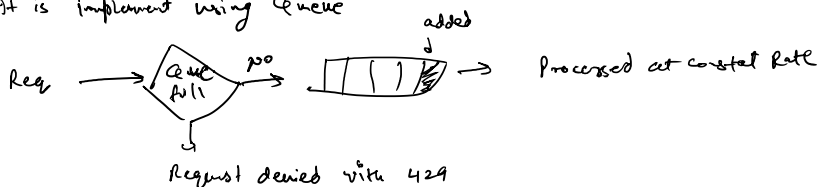
we can create rule:- Per user, Per minute ^{capacity} 3 tokens, Post request, Refill = 2 tokens

{ counter 3 }
 Req → { counter 2 Timestamp }
 Req → { counter 1 Timestamp }
 Req → { counter 0 Timestamp }
 Req → { Request denied } Return 429
 Req (after 2 min) → { counter 2 Timestamp }

10:1 → 0
 10:2 → 0 + 2
 10:3 → 2 + 2 = 4 but capacity 3 so I will overflow so 3 will be in bucket



It is implement using Queue

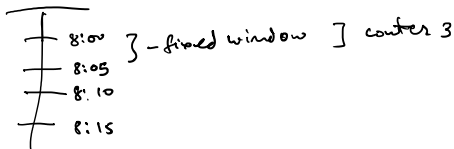


Disadvantage:-

ex. Amazon Drive in day traffic is Low and In Night its higher

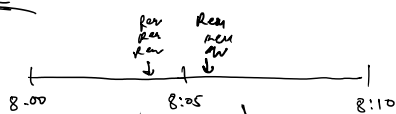
Fixed window Counter

It divide ^{24hr} timeframe into 2 or 5 min timeframe



Request work with Counter if counter is zero than it denies the Request

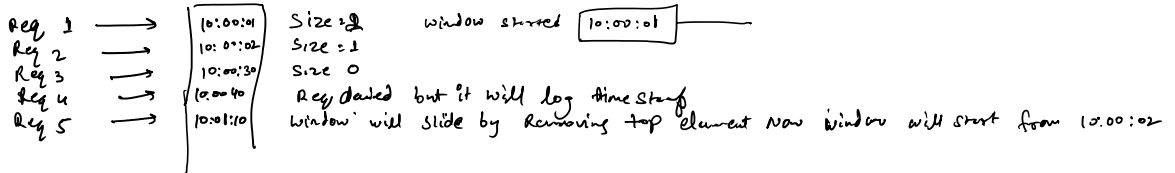
Disadvantage



Sliding window Log:-

lets say we have configuration 3 Req/min

we log timestamp

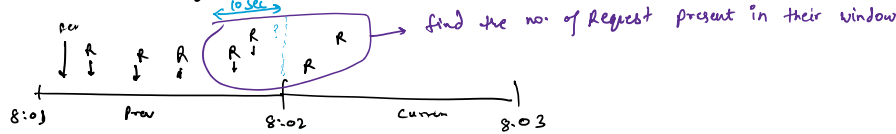


Disadvantage :- Even we are deny the Request, we are logging the Timestamp

Sliding Window Counter :-

- It provides 2 functionalities
- 1) fixed window counter
 - 2) sliding window log

Rule :- 1 min 5 Req allowed



$$= 2 + \left(\frac{6}{60} \times 10 \right)$$

6 sec → 6 req
10 → $\frac{6}{60} \times 10$

$= 2 + \left(\frac{1}{6} \right)$ → this is trying to find out how many Request and check with present then compare with Rule

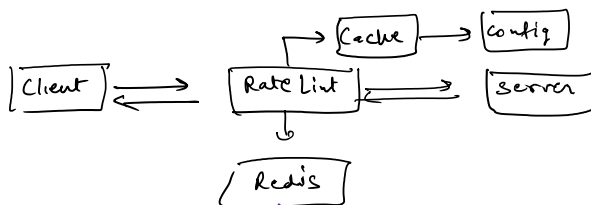
HLD

Components → we need counter so
 → disk X because its slow
 → Redis ✓
 we need config

There are two ways to Host Rate Limiter

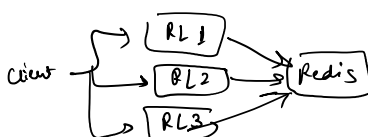


this can be put in Api gateway also



As soon as Server is up we will save Config files in Cache

→ this is centralized data store



Question :- Redis doesnot maintain Atomicity so if client send parallel Request to RL1, RL2 both will Read counter 2 from bucket How you maintain Atomicity?

Ans:- There are already solution present which brings atomicity to Redis but there will be latency